

STAT – 614: Applied Statistics
Spring Term 2245 *ONLINE*

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Office Hours:	MW: 11:00 AM – 3:50PM	HLC-2212
	TR: 2:00 PM – 3:50PM	HLC-2212
	F: By Appt on Zoom	

Course Material

Text:

Probability and Statistics for Engineering and the Sciences, Devore, 9th 2015. Cengage Learning.

ISBN: 9781305251809

Note: I will not require that any students in this course purchase the text listed above; however, periodically during the term problems may be selected from the text to work on outside of class.

Objectives

This course covers basic statistical concepts and techniques including descriptive statistics, probability, inference, and quality control (time permitting). The statistical package Minitab will be used to reinforce these techniques. The focus of this course is on statistical applications and quality improvement in engineering.

Office Hours

In addition to posting examples and content videos in MyCourses each week I will be available through Zoom to answer student questions on the material. The primary times that I will be on are listed for the individual sections above. Beyond those times feel free to email me and setup additional times to meet through Zoom.

Homework

Homework will be assigned periodically throughout the term; each assignment will be due on Thursday by 11:59PM the week after it is assigned unless directed otherwise. Your assignments should be PDF format and posted to the corresponding dropbox in MyCourses. Late assignments, or assignments turned into me by email will not be evaluated for a grade for any reason unless permission to do so is granted in advance. Homework assignments are expected to be typed, formatted appropriately and contain correct symbols and equations. If you are unsure of how to do this, please do not hesitate to contact me prior to the due date of any assignment.

Course Project

At the end of the term, you will be expected to submit a project where you will be analyzing a dataset of your choosing, consisting of real-world data. The source of your dataset must be cited in your project as well as any other sites, textbooks or miscellaneous sources. Datasets taken from statistics textbooks are not acceptable. To ensure a proper topic for your project is selected, a project proposal must be submitted prior to the end of week 8 of the term. Failure to submit a project proposal will result in a deduction of 10% from your final project grade.

Exams

This term there will be a midterm and final exam (during Finals Week). There will be a range of available dates to take each exam that will be provided prior to the corresponding assignment; however, when you begin the exam there is an enforced time limit of 180 minutes (3 hours). Exam submissions that go beyond the time limit are subject to late penalties based on the length of upload delinquency. You may use your notes and/or textbook for the exams; however, the exams are not group assignments and any individuals who work collaboratively on the exams will receive a failing grade. Exams should be word processed and formatted clearly including proper symbols and equation notation. Exam files should be formatted as PDF and uploaded to the dropbox in MyCourses prior to the due date. Exams that are not uploaded to the dropbox will not be evaluated unless accommodation is agreed upon prior to the time of the exam.

Honorlock:

Honorlock is a tool which will proctor your online exams and quizzes this semester. Honorlock is an online proctoring service that allows you to take your online exam virtually. You DO NOT need to create an account or schedule an appointment in advance. Honorlock is available 24/7. All that is required to use this service is a computer, a working webcam/microphone, your ID, and a stable internet connection. For more information about using Honorlock at RIT, review [Completing an AI-proctored Online Exam with Honorlock \(Student Guide\)](#). Exam requirements will be communicated prior to each exam. Honorlock will also be used to verify the identity of students taking an online version of this course.

Final Grade Determination

Homework:	25%
Project:	15%
Midterm:	30%
Final Exam:	30%
TOTAL:	100%

Final Grade Scale

93+	A
90 - 92	A-
87 - 89	B+
83 - 86	B
80 - 82	B-
77 - 79	C+
73 - 76	C
70 - 72	C-
60 - 69	D
Otherwise	F

Final Exam Conflict Policy:

The Registrar's Office will provide the final examination schedule no later than the first day of each term, and make it available to the entire RIT community. Instructors may not change the official date and time of the exam. In extraordinary circumstances, instructors may request their department head for a change in time. The department head will work with the college scheduling officer in an attempt to change the time of the exam. The decision of the department head shall be final.

In case of conflict where the student has two finals scheduled at the same time, service course examinations will take precedence over home department examinations. If both examinations are service course examinations, the class with the larger enrollment will have precedence.

Students have the right (if they wish) not to take three or more final exams in one day. In a case where the student does have three or more finals scheduled on the same day, service course examinations will take precedence over home department course examinations. If two or more of the examinations are in the home department, the department head will resolve the issue. If two or more examinations are service course examinations, the class with the larger enrollment will have precedence over the others.

In all cases by the last day of the 10th week of classes during fall or spring term or in the case of summer term or other sessions less than 16 weeks by the last day of 2/3 of the session, the student should submit a written request for rescheduling to the head of the home department, with a copy to the instructor being asked to provide the rescheduled examination. By the last day of the 12th week of classes the department head will, after consultation with the parties involved, notify the student of the date of the rescheduled examination. The decision of the department head shall be considered to be final.

If the instructor chooses not to give a formal final examination, it is the expectation that the instructor will treat the exam week as a full component of the academic term. During this exam week, appropriate educational activities should be scheduled, including the opportunity for students to benefit from the instructor's professional counsel.

Section 504 of the Rehabilitation Act

The Rehabilitation Act of 1973, reaffirmed by the 1992 Americans with Disabilities Act (ADA), was created to protect the rights of people with disabilities. Section 504 of the Rehabilitation Act specifically applies to colleges and universities and can have a direct impact on our day-to-day activities. The information below, "The Facts about Section 504 in the Classroom," describes in more detail the law, how it impacts the classroom, modifications that can be made to comply with the law, and resources available to assist in understanding the law. Please take the time to read this fact sheet carefully. If you need further information or have questions, contact Susan Ackerman, Disability Services Director, at extension 5-6988 or by e-mail at smacst@rit.edu.

It is every faculty member's responsibility to become familiar with the law and to provide reasonable accommodations. More importantly, we want to continue to do everything reasonable to assist our many talented and skilled persons with disabilities in reaching their potential. The statement below has been developed to reach out to students with different academic needs and should be read by faculty to their classes during the first week of the semester.

"RIT is committed to providing reasonable accommodations to students with disabilities. If you would like to request accommodations such as special seating or testing modifications due to a disability, please contact the Disability Services Office. It is located in the Student Alumni Union, Room 1150; the Web site is www.rit.edu/dso. After you receive accommodation approval, it is imperative that you see me during office hours so that we can work out whatever arrangement is necessary."

Disability Services Director: Catherine Lewis, SAU-1155, cbldso@rit.edu

Academic Accommodations Coordinator: Alyson Jones, SAU-1150, aljdco@rit.edu

Staff Assistant: Lindsay Shepard, SAU-1150, lcsdso@rit.edu

Disability Liaisons:

Joanne Roets, GCCIS – 5-4779; jxrgla@rit.edu; 2123 Golisano Hall

Fredda Bishop, KGCoe – 5-4595; flbeen@rit.edu; 2203 J. Gleason Hall

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C06.0 POLICY PROHIBITING DISCRIMINATION AND HARASSMENT

Scope: All members of the RIT Community shall comply with this Policy. This Policy is considered part of the conditions of employment, enrollment, or participation in RIT sponsored events or programs.

I. POLICY STATEMENT

The RIT community is committed to a diverse and dynamic learning, working, and living environment. It is committed to an environment which encourages, promotes and protects free inquiry and free expression. Members of the RIT community have the right to hold, express vigorously, defend and openly promote their ideas and opinions. The RIT Policy Prohibiting Discrimination and Harassment is not intended to restrict freedom of speech or any form of artistic or visual expression. The policy is also not intended to restrict discussion and debate in the classroom or academic forum. Protecting these values, however, does not include protecting acts of discrimination or harassment.

RIT will not discriminate in terms and conditions of employment, admission, and participation in programs or residential life. It prohibits discrimination and harassment on campus, or at any RIT activities off campus, by its administrators, faculty, staff, students and student organizations, and external organizations and individuals in their operations with RIT.

The initiation of an investigation of a potential violation of C6.0 precludes an individual from later requesting the use of policy C6.1 to investigate the same issue. In addition, once a C6.0 investigation is initiated, it must be fully investigated. Confidentiality will be maintained as described in the definition section of this policy.

Making an intentionally false charge of discrimination or harassment or retaliating against someone who has made a charge is as serious an offense as discrimination and harassment and is prohibited. Nothing in this policy relieves RIT of the obligation of adhering to federal, state, and local laws.

Week	M	T	W	R	F
1 1/13/2025	Basic Data Description/ Numerical and Graphical Descriptions				
2 1/20/2025	Probability				
3 1/27/2025	Discrete Probability Distributions/Discrete Random Variables (DRV)				
4 2/3/2025	Continuous Probability Distributions/Continuous Random Variables (CRV)				
5 2/10/2025	Sampling Distributions/ Estimation using Point Estimates				
6 2/17/2025	Interval Estimates/ Confidence Intervals (One Sample)				
7 2/24/2025	Review for Midterm Exam				
8 3/3/2025	<u>Midterm Exam Week</u> *Project proposal due Friday by 11:59PM				
9 3/10/2025	<u>Spring Break Week No Class</u>				
10 3/17/2025	One Sample Hypothesis Testing				
11 3/24/2025	Two Sample Inference				
12 3/31/2025	Two Sample Inference Cont.				
13 4/7/2025	Analysis of Variance (ANOVA)				
14 4/14/2025	Regression Analysis				
15 4/21/2025	Chi-Square Testing, Goodness of Fit & 2-Way Tables				
16 4/28/2025	Review for Final Exam/Start of Finals Week				

** This is only an approximate schedule Please refer to MyCourses for important course announcements.

Introduction to statistics Numeric/Graphical Descriptions

Population vs Sample:

Population: A well-defined collection of objects of interest to the study.

It is very difficult for statisticians to work with populations due to constraints with respect to time and/or resources.

Some examples of populations:

- All television sets produced from a manufacturer from 2020 to present.
- Patients who have been admitted to a local hospital for a specified range of dates
- All individuals who live within a defined geographic area (New York State, etc.)

Sample: A subset of individuals (units) from a population of interest. Preferably these samples should be selected such that the resulting subset of individuals is a good demographic representation of the population and those selected are done so at random to avoid potential bias. Any sample from a population should be large enough to achieve reliable estimates while avoiding being too large and increasing costs associated with sampling. The process and techniques associated with generalizing the behavior of the population from the behavior of the sample is referred to as inferential statistics.

Types of Data

Categorical Data: Often referred to as qualitative data, observations of this data-type can be classified or define an attribute of the observed individual.

Ordinal Categorical data - Observations that can be ordered or ranked such as year level of student, letter grade received on a quiz or patron rating of local hotel.

Nominal Categorical data - Observations of this form do not necessarily require a specific order such as handedness of a respondent, political affiliation, Method of transportation (auto, train, boat, etc.)

Numeric Data: Often referred to as quantitative data, observations of this type of data is typically the result of counting the number of units that have a specific characteristic of interest or a measurement associated with an individual.

Discrete data - Numeric data of this form take on specific values over a range (typically integer or whole numbers).

Discrete values will either be finite or countably infinite in number.

Number of defective parts produced over a 24 hour period.

Number of beds available at a local hospital

Number of individuals in support of decreasing taxes

Continuous data - Numeric data that can achieve any possible value in a range of possible values.* Possible values associated with continuous measurements are uncountably infinite.

Weight of a shipped package

Length of time, in seconds, to perform a task

Temperature of freshly produced part

*We will assume that any measurement/measurement system that will produce an observation for our sample can do so to an infinitesimally small degree allowing a differentiation between any 2 individuals in the sample.

Numerically Describing Data:

Numeric Summaries for Categorical/Qualitative data

- Categorical data whether nominal or ordinal can be tallied or counted and a percentage or proportion for each group can be calculated. The proportion of observations of any value observed can be calculated with the formula below.

$$\text{relative frequency of a value} = \frac{\text{number of times the value occurs}}{\text{number of observations in the data set}}$$

Numeric Summaries for Numeric/Quantitative data

- Quantitative data types can be summarized including measures of central tendency (average value) and/or variability (spread). Not all numeric summaries are created equal and depending on the behaviors observed in the sample one metric may be more appropriate than another.

Measurements for central tendency

- Mode** - The most repeated numeric value of the sample. There is no guarantee that a mode will exist, that there will be only one mode and the mode is not necessarily at the "center" of observed values.
- Median** - The "middle" value of the sample. The median of the sample will divide the number of observations in the sample in half based on the number of observations in the sample. The median is often referred to as the 50th percentile since 50% of the values will fall below the median.

The **sample median** is obtained by first ordering the n observations from smallest to largest (with any repeated values included so that every sample observation appears in the ordered list). Then,

$$\tilde{x} = \begin{cases} \text{The single middle value if } n \text{ is odd} & = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ ordered value} \\ \text{The average of the two middle values if } n \text{ is even} & = \text{average of } \left(\frac{n}{2}\right)^{\text{th}} \text{ and } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ ordered values} \end{cases}$$

- Mean** - The "center of mass" of the sample. The mean is the center of the sample with respect to the magnitude of the observations.

The **sample mean** $\bar{x}$ of observations $x_1, x_2, \dots, x_n$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

The numerator of $\bar{x}$ can be written more informally as $\sum x_i$, where the summation is over all sample observations.

*Due to the difference in definition of each of these measures of central tendency these values do not have to have the same magnitude but could be equal.

Measurements for variability or spread

- Range** - The difference between the minimum and maximum observed values of the sample.
- Interquartile Range** - Commonly abbreviated as IQR this is the difference between the lower quartile (25th percentile) and the upper quartile (75th percentile). The IQR represents the spread of the central 50% of the collected sample allowing for the removal of potential unusually small or large observations.
- Variance** - Formulaically the sample variance is defined by the following:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{S_{xx}}{n - 1}$$

Notice in this formula that each of the individual deviations from the mean are squared prior to summation as a means to counteract the positive and negative differences cancelling each other out leading to a net sum of zero. This creates an issue where the unit of measure for the sample variance is the square of the original unit of measure for the whole sample.

- Standard Deviation** - Like the variance, the standard deviation describes the average difference from an observation in the sample to the mean of the sample. Where standard deviation differs from variance is that the unit of measure of the standard deviation is the same as that of the sample allowing for an easier interpretation of the results.

$$s = \sqrt{s^2}$$

Graphically Describing Data:

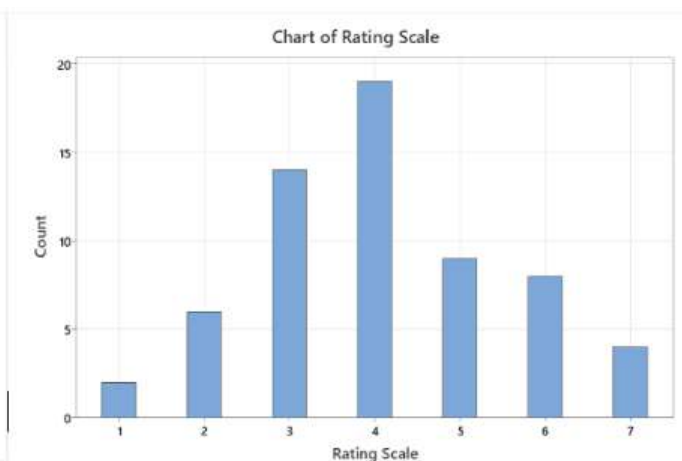
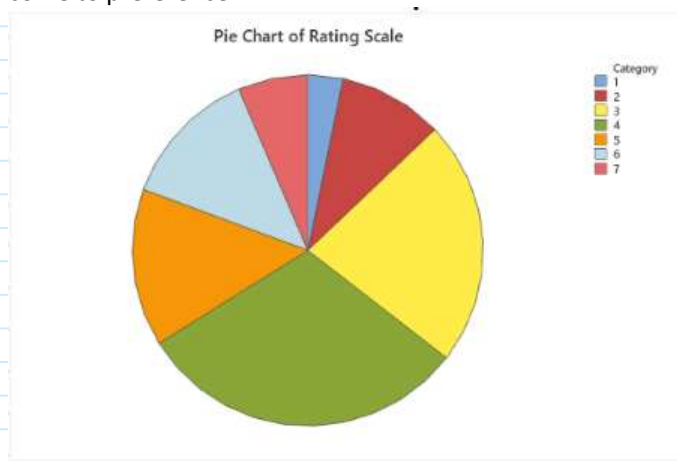
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Graphical Summaries for Categorical Data

- Pie Charts** - Visualizes the relative frequency (%) of the observations from a sample as a "slice of pie". The entire circle of

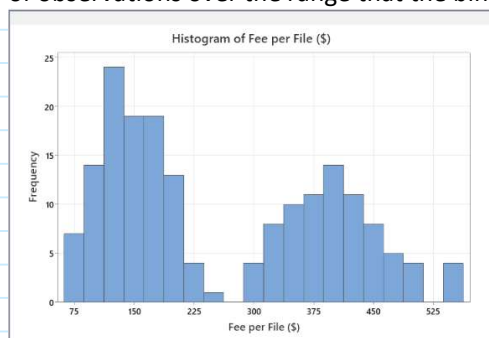
the pie chart represents 100% and allows for a quick comparison of each category observed.

- Bar Chart - Similar to Pie Charts; however uses rectangular bars to represent the frequency or percentage of the sample in each category. The same information can be gained from a bar chart as a pie chart. The use of one over the other can come to preference.

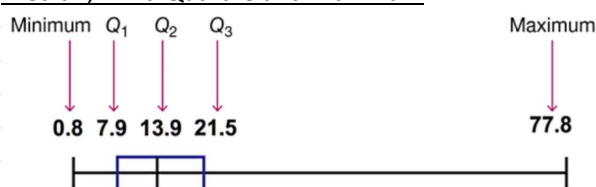


Graphical Summaries for Numeric Data

- Histogram - Similar to the Bar Chart seen above, however since the variable being plotted is numeric in nature the bars or "bins" associated with the histogram are allowed to touch each other. The absence of a bar in a histogram signifies a lack of observations over the range that the bin covers.



- Boxplot - The visual representation of the 5-number summary of the sample consisting of the Minimum, First Quartile, Median, Third Quartile and Maximum

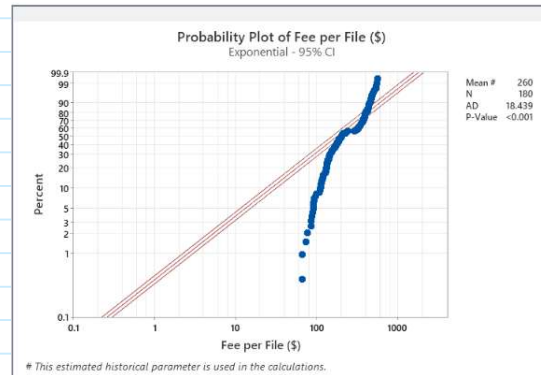
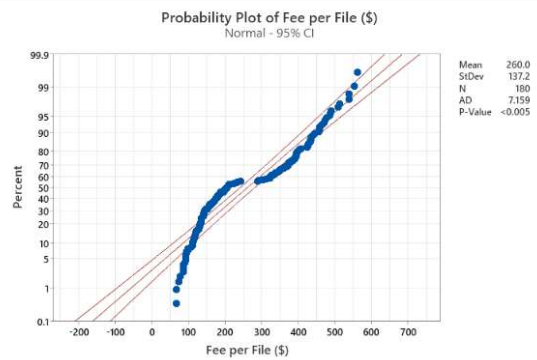


The information in a boxplot can be oriented horizontally or vertically, the choice is arbitrary and does not affect the interpretation of the information provided. Boxplots may be modified to show the presence of potential outliers in the sample. Any observation that is more than 1.5(IQR) away from the median value would be classified as a mild outlier. Any observation beyond 3(IQR) would be classified as an extreme outlier.



Due to the somewhat compact nature of the boxplot it is also fairly easy to create comparative boxplots while maintaining a fairly high level of readability.

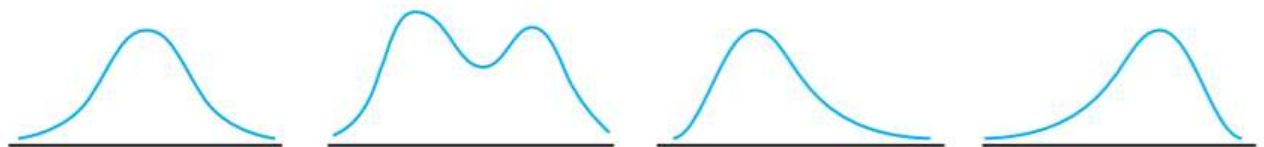
- Probability Plot - Useful to determine the likely distribution the observations from a sample are likely to have originated from, the most common is the Normal Probability Plot.



- Time Series Plot - Can only be constructed from data that changes over time. The time order of the observations must be known in order to construct and correctly interpret any time series plot.
- Scatter plot - Used in Bivariate (2 numeric variable) analysis to describe the nature of the trend between 2 numeric variables.

When we look at graphical displays of numeric data in the univariate cases we initially want to focus on 3 properties:

- Central location - Where does the main peak of the distribution fall?
- Shape - How many peaks in total does the distribution have? Is there only one? Are there many? Does the curve appear somewhat symmetric or is it skewed in some direction?



- Spread - How much variability does the distribution have?

Probability and Random Variables

Chance Experiment - An experiment that is run where the outcome of the experiment is completely random or left to chance.

Sample Space, S - The collection of all possible outcomes from a chance experiment.

Example: Our chance experiment is flipping a single coin

$$S = \{H, T\}$$

Example: Our chance experiment is Rolling a 6 sided die

$$S = \{1, 2, 3, 4, 5, 6\}$$

*When listing the elements of the sample space be sure to list all possible outcomes even if they are similar to outcomes previously listed.

Example: Our chance experiment is flipping 2 coins

$S = \{HH, TT, HT\}$ - *most people would stop here but there is one more outcome not listed.*

$S = \{HH, TT, HT, TH\}$ - *This is the completed sample space for this problem.*

Calculating the size of the sample space:

Method 1 - " n " identical objects with " k " individual outcomes

$$\text{Size of } S = k^n$$

Example: Our chance experiment is flipping 2 coins:

of elements = $(2)^2 = 4$ which is an indication that we would be missing an element in the sample space if we stop after the first different side observation.

Method 2 - " n " objects with " k " outcomes and " m " objects with " p " outcomes

$$\text{Size of } S = k^n p^m$$

Example: Our chance experiment is randomly selecting 3 digits from 0-9 and selecting 2 letters at random.

$$\text{\# of elements} = (10)^3 (26)^2 = 676000$$

*The above method can be adapted for any number of different types of

items used. An alternative way of handling sample size calculations is to use permutations and combinations.

$$P_r^n = \frac{n!}{(n-r)!} \text{ (for a subset of number } r\text{)}$$

$$C_r^n = \frac{n!}{r!(n-r)!}$$

Event - A subset of the sample space that contains outcomes of interest to the researcher.

Example: Suppose we have the experiment where we are flipping 3 coins.

Define event A as the event where we flip exactly 1 head.

$$A = \{HTT, THT, TTH\}$$

Define event B as the event where we flip 2 heads and Coin 1 must be heads.

$$B = \{HHT, HTH\}$$

Simple event - An event from the sample space that has exactly one outcome.

Null Event, $\emptyset$ - An event that contains no outcomes.

Probability (definition 1):

$$P(A) = \frac{\text{Outcomes in event } A}{\text{Size of sample space}}$$

- $0 \leq P(A) \leq 1$
- $P(S) = 1$
- $P(\emptyset) = 0$

Impossible Event - An event in the sample space with $P(A) = 0$; this event will never happen in the experiment.

Certain Event - An event in the sample space with $P(A) = 1$; this event is guaranteed to happen every time the experiment is run.

Complement of an event A - All outcomes from S that are not currently part of the event in question. Written as : $\bar{A}$, A' , A^{-1} , A^c

- $A + A^c = S$
- $P(A) + P(A)^c = P(S) = 1$
- $P(A) = 1 - P(A)^c$
- $P(A)^c = 1 - P(A)$

Intersection of Events - All elements that are contained in two events at the same time.

$$A \cap B$$

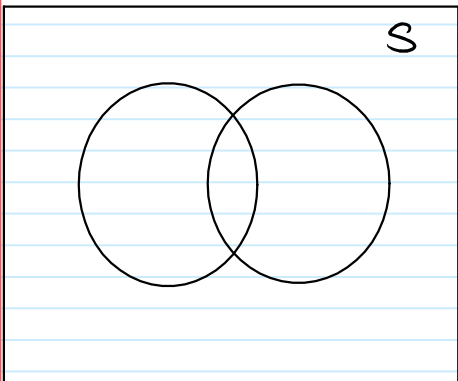
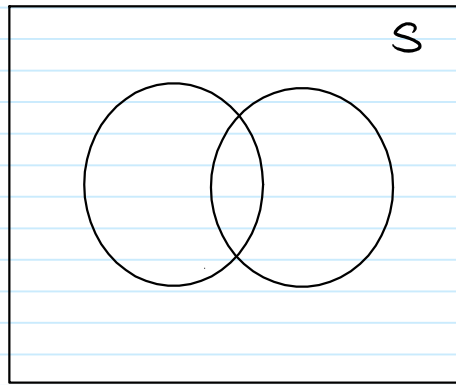
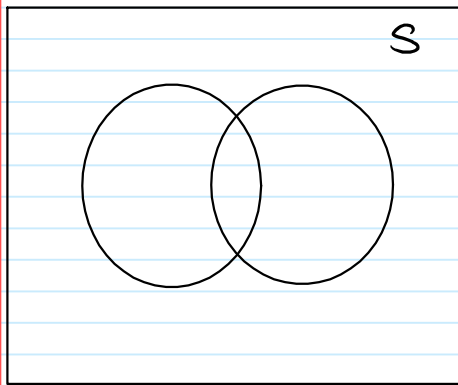
Union of Events - All elements that are contained in the first event, or the second event or both events at the same time (inclusive OR).

$$A \cup B$$

Venn Diagram: A way to visually represent the events (and probabilities of) within a sample space. The overall rectangular area labeled S represents all possible elements of the sample space. The areas bounded within the overall area represents the elements that are contained within various events that are defined.

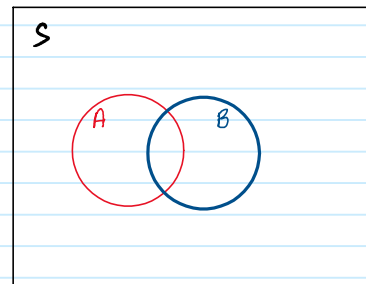
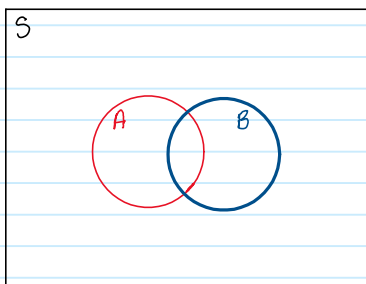
If the Venn diagram is drawn to scale we can calculate the probabilities of events happening with:

$$P(A) = \frac{\text{Area}(A)}{\text{Area}(S)}$$



De Morgan's Law: If we take the complement of the union between events it is the same as taking the intersection between the complements of each event. Alternatively, if we take the complement of the intersection between events it is the same as the union between the individual complements.

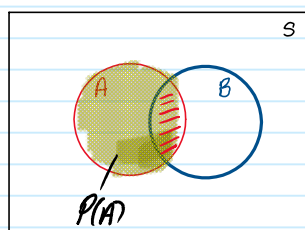
$$P(A \cap B)^c = P(A^c \cup B^c) \text{ and } P(A \cup B)^c = P(A^c \cap B^c)$$



Conditional Probability - When the probability that an event may happen depends on information about another event that has previously happened.

$$P(A|B)$$

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(B|A) = \frac{P(A \cap B)}{P(A)}$



General Multiplication rule for Probability:

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

General Addition Rule for Probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

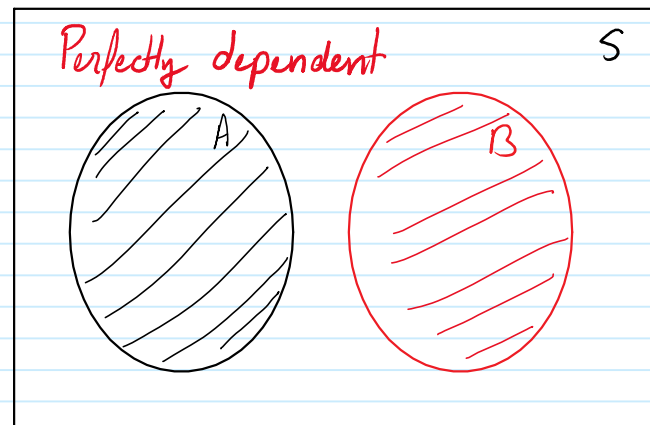
Disjoint events (Mutually Exclusive) - Events that share no outcomes in common. The intersection between these events will be null.

$$P(A \cap B) = 0$$

$$P(A|B) = P(B|A) = 0$$

Addition rules for **Disjoint** events:

$$P(A \cup B) = P(A) + P(B)$$



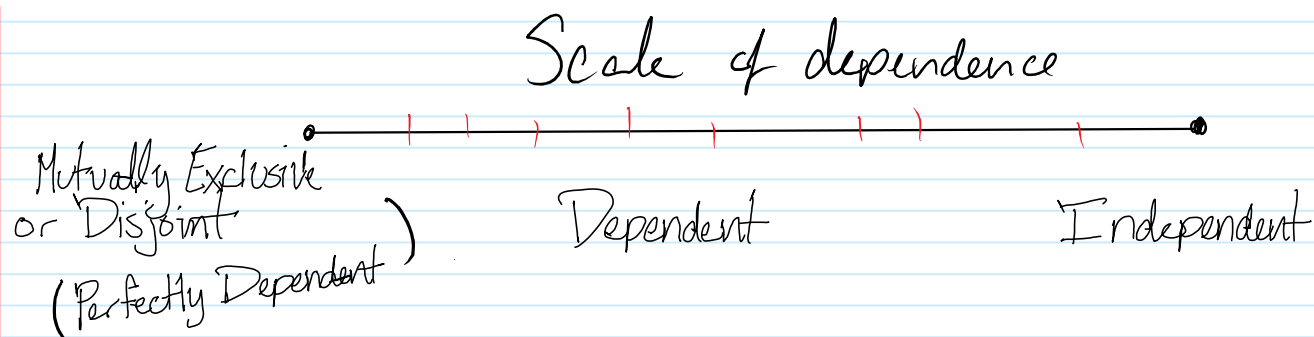
Independence of events: When the knowledge of an event already happening does gives no additional information about a second event that may or may not happen.

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Multiplication rule for **independent** events

$$P(A \cap B) = P(A|B)P(B) = P(A)P(B)$$



Total Probability: Assume in a system there are several possible paths you can follow where each path is disjoint of any other path. If each path can potentially end in an event, the total probability of that event is given by:

$$P(A) = \sum P(A|B_i)P(B_i)$$

You are a commuter student who lives in Brighton and you have 2 possible routes you take to campus. If you take route 1 you have a 70% chance to arrive on campus on time. If you take route 2 you have a 59% chance of arriving on time. If you decide to choose route 1 35% of the time, what is the overall probability that on a day where you are commuting to campus that you will arrive on time?

Bayes' Theorem: Conditional probability where the time dependency of the condition has been flipped

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum P(A|B_i)P(B_i)} = \frac{P(A \cap B_i)}{P(A)}$$

You arrived at campus on time, what is the probability you chose route 1 to get here?

Probability definition 2: Experimental/Empirical approach

Classical approaches to probability fail if either the event or the Sample Space are not clearly defined or infinite in size. Empirical probability is a method to estimate the probability of an event based on the results gathered during a sample experiment.

$$P(X = x) \approx \frac{\text{\#successful trials}}{\text{Total \# of trials}}$$

Law of large numbers - as the total number of trials of the experiment approaches infinite, the probability under the empirical definition will approach the true probability of the event of interest.

Example: In a random sample of 259 students, each individual was asked if they owned a tablet and if they owned a smart phone.

	Owens Smart Phone	Does NOT own a Smart Phone	Total
Owens Tablet	79	3	82
Does NOT own a tablet	161	16	177
Total	240	19	259

Based on this table, what is the probability that a student owns a smartphone?

What is the approximate probability that a randomly selected individual will own a tablet and a smart phone?

What is the approximate probability that a randomly selected individual will not have a tablet or a smart phone? (neither device)

Random Variable - a numeric variable whose value represents the outcome of a chance experiment.

**Random variables can be Discrete or Continuous in nature. For discrete variables the Probability Mass Function, pmf, describes the probability that the variable will take on a specific value of that variable. For continuous variables this is referred to as the Probability Density Function or pdf.

EX: Experiment to flip 3 coins

RV: X # of heads observed

X	P(X=x)
0	
1	
2	
3	

For any DRV:

- $0 \leq P(X = x) \leq 1$

- $\sum P(X = x) = 1$

*The Cumulative Distribution Function, CDF, represents the probability that the observed value of the random variable would be less than or equal to a chosen value of the variable. The CDF of a variable x is represented with the notation $F(x)$

For any CRV:

- $f(x) \geq 0$ for all x where X is defined.
- $\int f(x)dx = 1$ the total area under a probability density function must always equal 1!

Is $f(x) = x^2$ a valid density function over $0 \leq X \leq 1$?

Is $f(x) = \frac{1}{2}x$ a valid density function over $0 \leq X \leq 2$?

What is $P(\frac{1}{4} \leq X \leq \frac{3}{2})$?

Expected Values and Special Discrete Distributions

Expected value of DRV's

The expected value is the long term or long run average value of the random variable. This average value may or may not be a value the variable can actually achieve.

For DRV:

$$E(X) = \mu_x = \sum xP(X = x)$$

For CRV:

$$E(X) = \mu_x = \int xf(x)dx$$

Example for 3 coin flip example from last lecture:

For the random experiment of flipping 3 coins, how many coins do we expect to land heads up?

X	P(X=x)
0	
1	
2	
3	

For the density function $f(x) = 1/2 x$ over $0 \leq X \leq 2$, calculate the mean value of X .

Variance of Random Variables:

The variance of a RV measures the variables tendency to deviate from its expected value.

For DRV:

$$V(X) = \sigma_x^2 = \sum [x^2 P(X = x)] - \mu_x^2$$

For CRV:

$$V(X) = \sigma_x^2 = \int (x - \mu_x)^2 f(x) dx$$

**Variance of ANY random variable should be positive if the variable takes on at least 2 unique values.

Example using the 3 coin flip experiment from above:

What is the variance in the number of heads that are flipped on 3 coins?

How variable is X defined by the density $f_{(x)} = 1/2 x$ over $0 \leq X \leq 2$?

Special DRV Distributions:

Remember for any DRV:

- $0 \leq P(X = x) \leq 1$
- $\sum P(X = x) = 1$

1 - Binomial Distribution

Binomial Distribution:

- Independent Bernoulli trials
- Number of trials, n, is fixed at start of experiment
- Number of successes, x, is between 0 and n
- P(success) is constant for all individual trials

$$P(X = x) = \binom{n}{x} P(s)^x [1 - P(s)]^{n-x} \text{ where } \binom{n}{x} = n^C x = \frac{n!}{x! (n-x)!}$$

$$E(X) = nP(s)$$

$$V(X) = nP(s)[1 - P(s)]$$

2 - Hyper Geometric Distribution

- Variable X: # of successes present in Collected Sample.
- fixed Bernoulli trials n (non Independent)
- Population size, N, finite so P(s) Non Constant
- a: # of successes in Population.

$$P(X = x) = \frac{\binom{a}{x} \binom{N-a}{n-x}}{\binom{N}{n}}$$

$$E(X) = n \left(\frac{a}{N} \right)$$

$$V(X) = n \left(\frac{a}{N} \right) \left(1 - \frac{a}{N} \right) \left(\frac{N-n}{N-1} \right)$$

3 - Poisson Distribution

- Variable X: # successes or arrival of successes over change in time/position
- λ – rate of successes (constant)
- $x \geq 0$ over Δt

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$E(X) = \lambda$$

$$V(X) = \lambda$$

4 - Geometric Distribution

- Variable X: # of trials performed before first successful outcome
- $P(s)$ = Probability of success of individual trials (constant)
- Independent Bernoulli trials
- 1st success must happen on last trial of experiment.

$$P(X = x) = P(s)[1 - P(s)]^x$$

$$E(X) = \frac{1}{P(s)} \quad * \text{Expected total number of trials}$$

$$E(X) = \frac{[1 - P(s)]}{P(s)} \quad ** \text{Expected total number of failures}$$

$$V(X) = \frac{[1 - P(s)]}{P(s)^2}$$

5 - Negative Binomial Distribution

- Variable X: # of trials needed to achieve r number of successes
- $P(s)$ = Probability of success of individual trials (constant)

- Independent Bernoulli trials
- r^{th} success must happen on last trial of experiment.

$$P(X = x) = \binom{x-1}{r-1} P(s)^r [1 - P(s)]^{x-r}$$

$$E(X) = \frac{r}{P(s)} * \text{Expected total number of trials}$$

$$E(X) = \frac{r[1 - P(s)]}{P(s)} ** \text{Expected total number of failures}$$

$$V(X) = \frac{r[1 - P(s)]}{P(s)^2}$$

Multinomial Distribution

- Like the Binomial distribution but more than 2 potential outcomes that are Disjoint of each other.
- Trials of the experiment are independent of each other and
- $P(A), P(B)$ and $P(C)$ are constant over the experiment
- $a + b + c = n$

$$P(A = a, B = b, C = c) = \frac{n!}{a! b! c!} P(A)^a P(B)^b P(C)^c$$

No	Question	Yes
Go to 2	Fixed Population size (finite)?	Hyper Geometric
Go to 3	Rate over time or distance with fixed time interval?	Poisson
Go to 4	Fixed number of total trials?	Binomial
Go to 5	Experiment ENDS after first success?	Geometric
World Ends!	Experiment requires multiple successes before ending	Negative Binomial

Continuous Random Variable Distributions

As a reminder for any CRV:

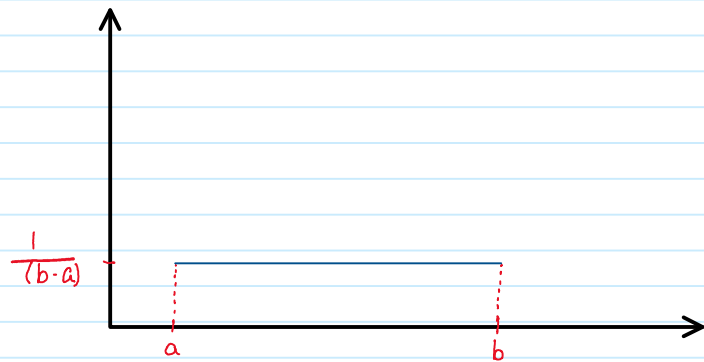
- $f(x) \geq 0$ for all x where X is defined.
- $\int f(x)dx = 1$ the total area under a probability density function must always equal 1!

1.) Uniform Distribution: Density function remains constant over range of X defined:

$$f(X) = \frac{1}{b-a} \text{ for } X \text{ between } a \text{ and } b; 0 \text{ otherwise}$$

$$\mu_x = \frac{1}{2}(b+a)$$

$$\sigma_x^2 = \frac{1}{12}(b-a)^2$$



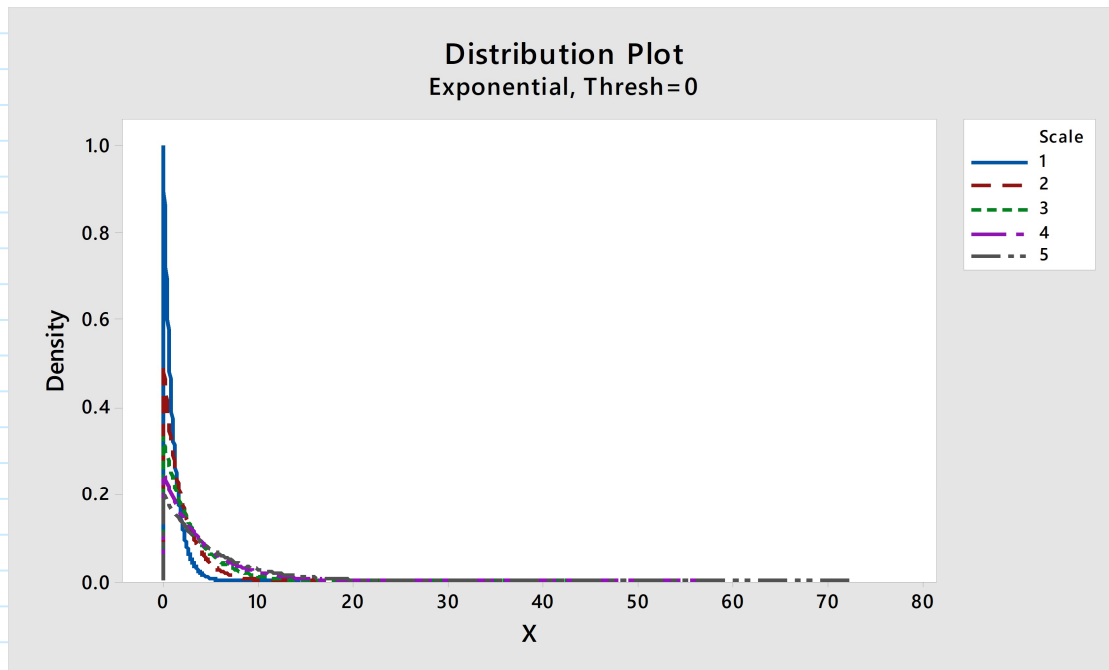
2.) Exponential Distribution:

$$f(x) = \lambda e^{-\lambda x}$$

OR

For $X \geq 0$

$$f(x) = \frac{1}{\beta} e^{-\frac{x}{\beta}}$$



$$\mu_x = \frac{1}{\lambda} = \beta$$

$$\sigma_x^2 = 1/\lambda^2 = \beta^2$$

Poisson vs Exponential Distributions:

Poisson (DRV) - variable represents number of successes to happen over an interval of fixed length.

Exponential (CRV) - variable represents time between 2 Poisson successes.

3.) Normal Distribution (Gaussian):

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } -\infty \leq X \leq \infty$$

$$\mu_x = \mu$$

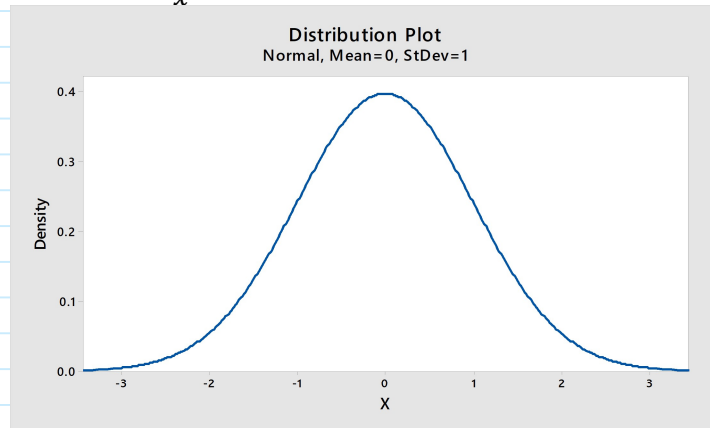
$$\sigma_x^2 = \sigma^2$$

Special Case of the Normal Distribution - Standard Normal or z distribution

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } -\infty \leq X \leq \infty$$

$$\mu_x = 0$$

$$\sigma_x^2 = 1$$



*To make working with the normal distribution easier we typically use tables that will give us the area under the curve from $-\infty$ to a specified value of X , the downside is that most tables available reference the z distribution. To use the tables we will have to standardize the values from the original distribution.

$$z_i = \frac{\chi_i - \mu}{\sigma}$$

4.) Lognormal Distribution:

If we have a variable, W , that is known to have a normal distribution with mean of μ and a variance of σ^2 ; then the relationship given by: $X = e^W$ would produce a distribution for the variable X that is lognormal with density function given by:

$$f(x, \mu, \sigma) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} e^{-[\ln(x)-\mu]^2/(2\sigma^2)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

5.) Weibull Distribution:

For any random variable X which follows a Weibull distribution we can say that the density function will be of the form:

$$f(x, \alpha, \beta) = \frac{\alpha}{\beta} \left(\frac{x}{\alpha}\right)^{\beta-1} e^{-\left(\frac{x}{\alpha}\right)^\beta} \quad \text{for any } x > 0 \text{ with a scale}$$

parameter given by $\alpha > 0$ and a shape parameter given by $\beta > 0$

*The scale and shape parameters can be modified for the Weibull distribution to change the overall look of the curve dramatically.

Continuous Random Variable Distributions

As a reminder for any CRV:

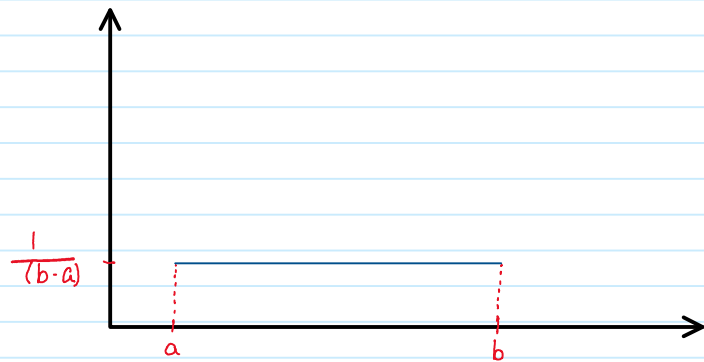
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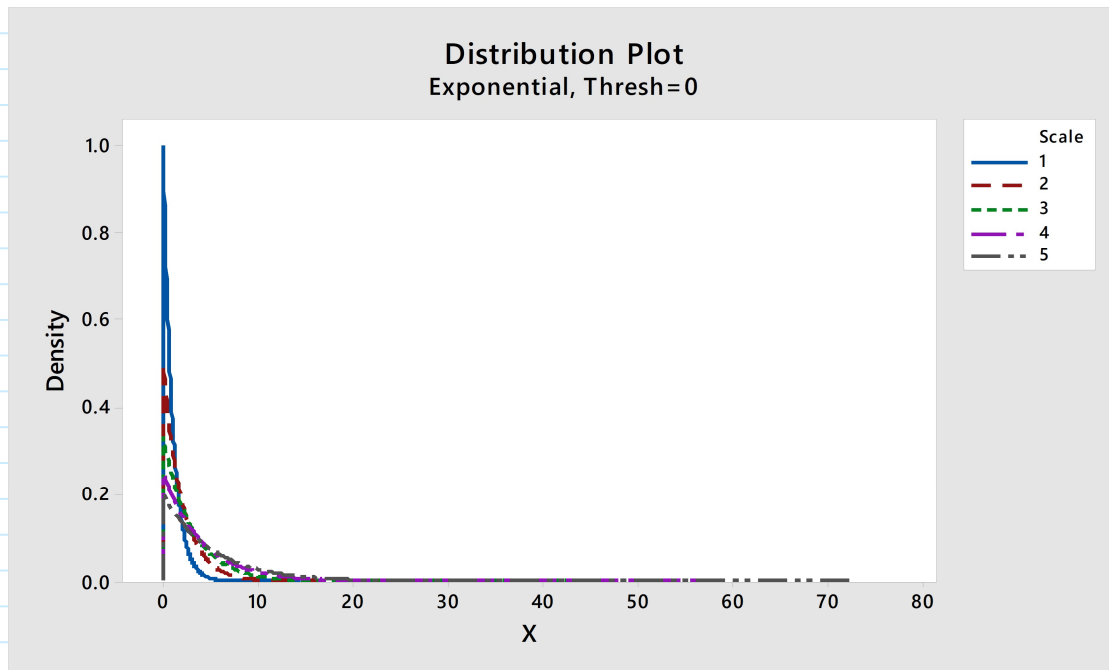
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OR

For $X \geq 0$

$$f(x) = \frac{1}{\beta} e^{-\frac{x}{\beta}}$$



$$\mu_x = \frac{1}{\lambda} = \beta$$

$$\sigma_x^2 = 1/\lambda^2 = \beta^2$$

Poisson vs Exponential Distributions:

Poisson (DRV) - variable represents number of successes to happen over an interval of fixed length.

Exponential (CRV) - variable represents time between 2 Poisson successes.

3.) Normal Distribution (Gaussian):

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } -\infty \leq X \leq \infty$$

$$\mu_x = \mu$$

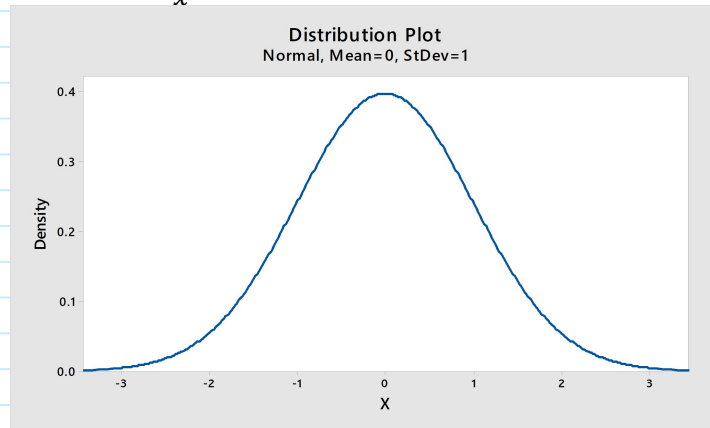
$$\sigma_x^2 = \sigma^2$$

Special Case of the Normal Distribution - Standard Normal or z distribution

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } -\infty \leq X \leq \infty$$

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$$\sigma_x^2 = 1$$



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For any random variable X which follows a Weibull distribution we can say that the density function will be of the form:

$$f(x, \alpha, \beta) = \frac{\alpha}{\beta^\alpha} (x)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^\alpha} \quad \text{for any } x > 0 \text{ with a shape parameter given by } \alpha > 0 \text{ and a scale parameter given by } \beta > 0$$

*The scale and shape parameters can be modified for the Weibull distribution to change the overall look of the curve dramatically.

For the Weibull distribution:

$$\mu_x = \beta \Gamma\left(1 + \frac{1}{\alpha}\right)$$

$$\sigma_x^2 = \beta^2 \left[\Gamma\left(1 + \frac{2}{\alpha}\right) - \left[\Gamma\left(1 + \frac{1}{\alpha}\right) \right]^2 \right]$$

Estimation - Point Estimates

Inferential statistics involves using a sample, and the statistics from that sample to estimate the value(s) of the parameters for the population. We need to do this because the population:

- May be of infinite size
- So large that the amount of resources required to observe all possible individuals is massive

And by comparison it is easier to select a random sample from that population.

The statistics that we calculate are random variables since the specific value the statistic takes on is likely to change from one sample to the next. This means that every statistic will have a probability distribution with an expected value and a variance. The distribution that is associated with any statistic is also known as the **sampling distribution** for that statistic.

The standard error is the standard deviation of the sampling distribution of a statistic.

The Central Limit Theorem:

PART 1 - If we sample from a known normal population with mean and standard deviation of μ and σ ;

We know the distribution of the sample mean, will also be normal with a mean μ and a standard error given by $\frac{\sigma}{\sqrt{n}}$ for ANY sample size n greater than 1.



PART 2 - If we sample from a random population (NON-normal) with mean and standard deviation of μ and σ ;

The distribution of the sample mean will be APPROXIMATELY normal with mean μ and a standard error given by $\frac{\sigma}{\sqrt{n}}$ for a sample size n that is "sufficiently large".

* $n \geq 30$

Part 3 - If we sample from a known Binomial Population with a population percentage of success given by p ;

The distribution of the sample proportion, $\hat{p}$, will be approximately normal with a mean p , and standard error given by $\sqrt{\frac{p(1-p)}{n}}$, for a sample size n that is "sufficiently large".

$$*np(1-p) \geq 10$$

Estimation

Point Estimation - We will use a single value (point) that is a reasonable value of the population parameter. In general the statistic that produces a value of the point estimate is referred to as a point estimator. We try to choose estimators that, on average, will be equal to the value they are estimating (unbiased).

This means that for an estimator, $\hat{\theta}$, which will be used to estimate the value of the population parameter Φ the following would be true:

$$E(\hat{\theta}) = \Phi$$

Above we saw that the expected value of the sample mean was equal to μ , the population mean, so we can say that the sample mean is an unbiased estimator of the population mean.

In general for the following parameters we would choose the point estimators:

Parameter	Estimator
μ	$\bar{X}$
σ	s
π	$\hat{p}$

Estimation - Interval Estimates (One Sample)

Point Estimate: An estimate for a parameter that is a single calculated value from a sample.

Good point estimates are unbiased and have little variability.

Interval Estimates: An estimate for a population parameter that uses a range of values we expect or are confident that the parameter's value will fall between.

A.K.A a "confidence Interval"

General Interval estimate method: (works for most intervals in this class)

$$\text{point estimate} \pm \text{Margin of Error}$$

where the Margin of Error is typically defined by:

$$\text{Margin of Error} = \text{Critical Value} * \text{Standard Error of Estimate}$$

Estimating the Population mean, μ , with one sample:

If we know the sampling distribution is approximately normal, and we know the true standard deviation from the population, we may estimate the mean with $(1 - \alpha)\%$ confidence with the following method:

$$\bar{X} \pm z_{(1-\alpha/2)}^* \left\{ \frac{\sigma}{\sqrt{n}} \right\}$$

However, if the true deviation is unknown, we will need to use a reasonable point estimate for the deviation. To do this we will use our sample standard deviation. This method will now have more than one source of variability and we will need to widen the interval bounds to account for the increase in variability. This will involve a change of critical value using a t-distribution instead of the standard normal distribution.

$$\bar{X} \pm t_{(v, \alpha/2)}^* \left\{ \frac{s}{\sqrt{n}} \right\}$$

Where the t-critical value is associated with degrees of freedom, v , given by $v = n - 1$.

Estimating the population deviation, σ , with one sample:

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with parameters μ and σ^2 . Then the rv

$$\frac{(n-1)s^2}{\sigma^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma^2}$$

has a chi-squared (χ^2) probability distribution with $n - 1$ df.

This means that unlike the methods above, our interval associated with the variance(deviation) will not be symmetric as it is with the mean.

For a one sample interval estimate for the true variance we have the following:

$$\frac{(n-1)s^2}{\chi^2_{(v, \alpha/2)}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{(v, 1-\alpha/2)}}$$

Where the critical values for the bounds are from the Chi-Square distribution with $n-1$ degrees of freedom. We can convert these bounds to represent an estimate for the standard deviation by square rooting both values leading to the bounds:

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{(v, \alpha/2)}}} \leq \sigma \leq \sqrt{\frac{(n-1)s^2}{\chi^2_{(v, 1-\alpha/2)}}}$$

Estimating the population proportion, π , with one sample:

Under the central limit theorem we saw that when sampling from a Binomial (Bernoulli) population of success/failure observations, the behavior of our sample proportion would be approximately normal under the large sample condition criteria. When this assumption is reasonable to make we can estimate the proportion of success in the entire population with the method:

$$\hat{p} \pm z^*_{(1-\alpha/2)} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Which involves using critical values from the standard normal distribution and degrees of freedom are not necessary.

Special note: Each of these methods listed assume a two-sided approach to the estimation of the parameter in question. We can also decide to estimate parameters using single sided estimate methods

We should also be careful interpreting the bounds of any interval estimate. The variable in this situation is the method selected and the bounds that it produces; the parameter we are estimating is not a variable and we need to interpret our bounds as such.

In general when we increase our level of confidence we increase the overall width of the interval estimate since we are choosing larger magnitudes for our critical values. If we need the interval to be narrower while maintaining a specific level of confidence we need to increase our sample size.

Approximately, an increase in the sample size by a factor of 4 will cut the width of the interval in half but this will significantly increase the cost to perform the study in question.

The Normal Test/Normal Probability Plot

Several of our interval estimate and testing methods have a potential need for populations that were sampled from to be normally distributed in order to argue the method or process selected will be valid. Rarely will the problem tell us that the population that has been sampled from will be normal so we need to apply special tools to help us make this inference.

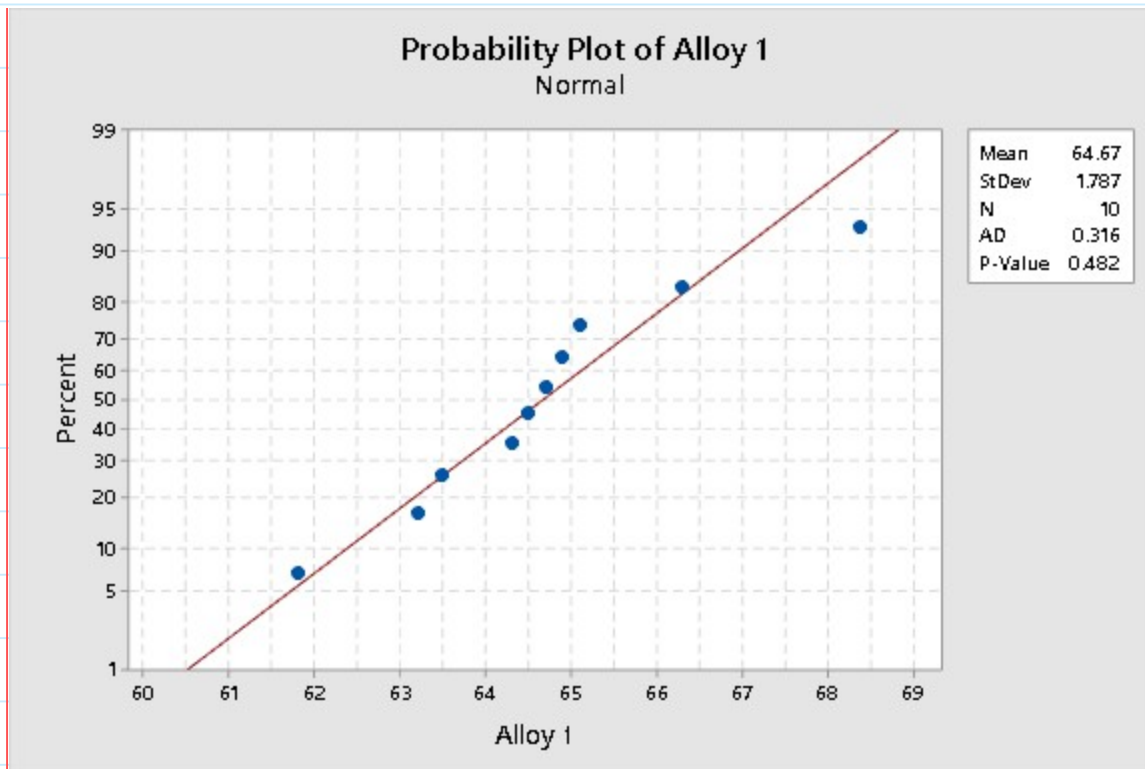
The Normal Test:

We can plot the sample data against a theoretical normal distribution with the same mean and standard deviation as our sample. The more linear the plot the more likely it is that our sample came from a normally distributed population. A test can also be constructed to determine just how close to normality our sample falls. This test was proposed by Theodore Wilbur Anderson and Donald A. Darling with the following interpreted Hypotheses:

H_0 : *The collected sample comes from a normally distributed population*

H_a : *The collected sample does NOT come from a normally distributed population*

The test statistic and P-value for this test can be found to the right side of the normal probability plot when the Normal Test command is used. An example is highlighted for the problem below:



Notice that the observations follow a relatively straight line for having only 10 observations in the sample and we have a high P-value (relative to α assuming a 5% level). This would lead us to the decision to fail to reject H_0 .

The conclusion for this test would then be: We do not have the evidence at a 5% level to claim that the measurements of hardness for alloy 1 are a sample that was not taken from a normally distributed population.

This conclusion is wordy but if we condense the statement down to the important bits we are saying that "The sample is not not normal" which is what we want if we need to make the assumption that the sample comes from a normally distributed population.

Inference - One Sample Hypothesis Testing

In the last section of notes we explored estimating the parameter of a population with point and interval estimates. We also introduced the concept of level of confidence in the estimates. Now we will turn our attention to inference about the population parameter(s) based on the test of hypotheses. For the purposes of this class we treat these hypotheses as potential claims about the parameter and we would like to know what claims, if any, are the most plausible.

Like the interval estimates previously discussed, hypothesis testing will give us an idea how the population is behaving; however, unlike intervals a test is performed to what is referred to as the **level of significance** of the test.

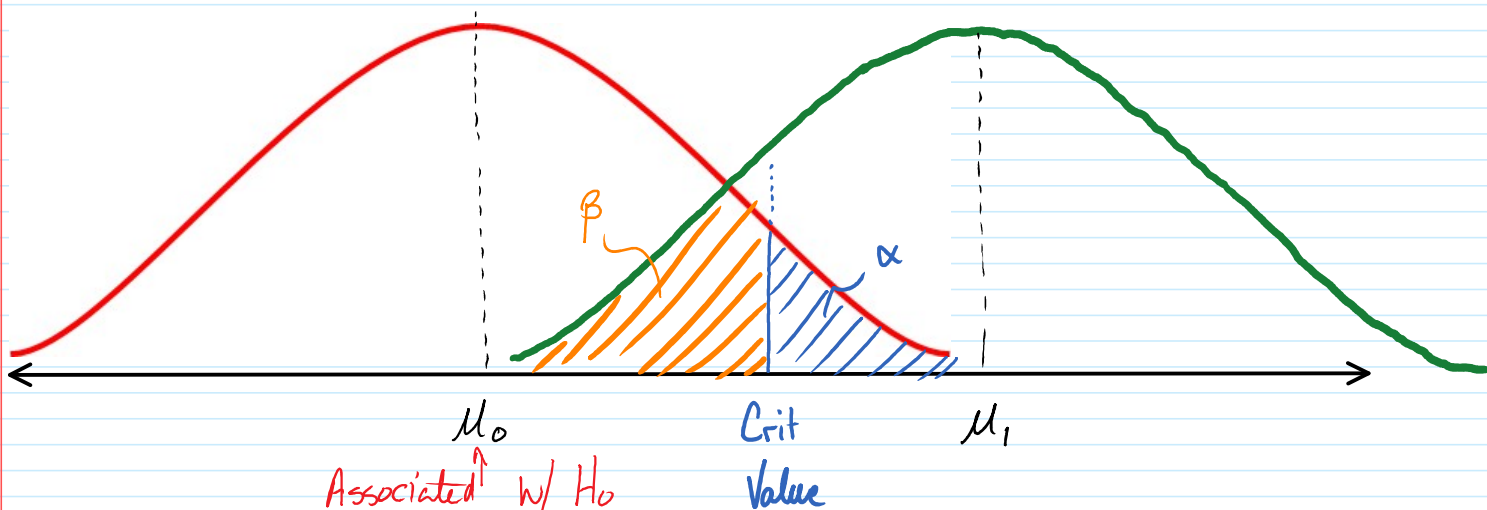
The 7 Step Hypothesis Testing Process:

1. Clearly define the parameter of interest. (value from the population that we are trying to draw inference on.)
Note: We do not assign any value to the parameter.
2. Supply/Formulate the Null Hypothesis (H_0)
Note: A statement about the parameter that is assumed true at the start of the testing process until shown otherwise. Will **ALWAYS** a statement of equality.
3. Supply an alternative hypothesis (H_a , H_1)
Note: This is a counter-claim against the statement of the null hypothesis. ($>$, $<$ or $\neq$)
4. Determine level of significance (α) of the test. Note: Use a default level of $\alpha = 0.05$ if it is not specified in the problem.
5. Perform ALL necessary calculations:
 - a. Start with the "4 questions"
 - o What kind of data are we working with?
 - o What type of parameter are we trying to estimate?
 - o How many of the parameter do we have? (how many samples)
 - o Are there any other assumptions we need to make or are allowed?
 - b. Calculation of Test Statistic
 - c. Calculation of P-value
 - d. Verifying the test validity (CLT)
6. Make a decision:
We can reject H_0 if the P-value $\leq \alpha$
OR
We can fail to reject H_0 if the P-value is $> \alpha$.
7. Write an informative conclusion in the context of the problem.
Ex.
If we reject H_0 : We have sufficient evidence at the 5% level of significance to support the claim that the true GPA of all RIT students is greater than 3.55.
If we fail to reject H_0 : We have insufficient evidence at the 5% level of significance to support the claim that the true GPA of all RIT students is greater than 3.55.

Errors in Hypothesis Testing:

True Nature of Universe		
	H_0 True	H_0 False
Reject H_0	No Error	Type II Error,

We believe Reject H_0 or Fail to Reject H_0	No Error	Type II Error, β
	Type I Error, α "Level of Sig"	No Error



* In general as $\alpha \rightarrow 0$, $\beta \rightarrow 1$
as $\beta \rightarrow 0$, $\alpha \rightarrow 1$

The P-value of a test is the probability we would have a future test statistic that is more extreme than our current statistic under the assumption that H_0 is true.

In the context of the hypothesis test, the P-value could also be considered the observed risk that we will commit a type I error, or if the null hypothesis were actually true the P-value would represent the probability that we are making an erroneous claim about the parameter.

Power of the hypothesis test ($1 - \beta$): the probability of successfully rejecting a false null hypothesis, or the probability of avoiding the type II error outcome.

Hypothesis testing about the mean of a population.

Assuming the distribution of the sample mean is approximately normal and that

the true standard deviation, σ , is known; we can define the test statistic for the one sample test on the mean as:

$$z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Where under the null hypothesis we are assuming $H_0: \mu = \mu_0$

The P-value for the statistic above would be the area under the standard normal curve that is "more extreme" than your current statistic assuming that H_0 is true. This means for a left tailed test we will calculate the area to the left of the current z statistic. For a right tailed test we will calculate the area to the right of the statistic. For a 2-tailed test we will first determine the side of the distribution that the statistic is on, then we will calculate the tail area and multiply by 2.

If we take the situation posed above and remove the assumption that the true standard deviation is unknown we can use the following method to calculate the test statistic based on the Student's t distribution:

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

Unlike the z test performed earlier, this t-test is associated with n-1 degrees of freedom. The behavior of the P-value and decisions are made exactly the same as they are for the z test.

Hypothesis testing about the proportion of a population.

When a sample is derived from Bernoulli type data (success/failure), provided CLT 3 applies to the problem, we may assume that the distribution of the sample proportion ($\hat{p}$) is approximated by a normal distribution. In this situation, we can generalize the one sample test statistic for the proportion using the z statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Previously in the case of working with interval estimates we estimated the standard error of the proportion with the sample proportion due to the lack of knowledge of the value for p (π). This is not the case for hypothesis testing since we need to operate under the assumptions that H_0 is true.

Hypothesis testing about the standard deviation of a population.

Building on the foundation established from interval estimates we know that when we divide the sum of square deviations from the mean by the population variance we get a χ^2 distribution with n-1 degrees of freedom. From this it is reasonable to say that the 1 sample test statistic for variance/standard deviation should be of the form:

$$\chi^2 = \frac{(n - 1)s^2}{\sigma^2}$$

$$\chi^2 = \frac{s^2}{\sigma^2}$$

Any decision reached as a result of performing a 2 sided hypothesis test will be consistent with the corresponding interval estimate performed at the complementary confidence level. One sided tests will correspond to the related upper or lower bound estimates of the parameters at the complementary level.

2-Sample Inference - Interval Estimates

As we saw previously when dealing with confidence intervals, we were interested in constructing bounds that we expected to contain the value of the parameter of interest to the study.

We will be conducting similar analyses when involving multiple samples; however, the method used to collect the samples must be taken into account.

Inferences concerning means

Independent vs. Paired (dependent) Sampling:

Paired sampling happens whenever you have a strong reason to link or pair observations from one distribution to observations from a second distribution. Situations where this is likely to occur are when you are observing the same individual at 2 points in time (before vs after) or using a similar type of item from different locations (same part type purchased). You can also link observations based on a relationship that exists between the individuals that are being observed. For each of these case types the number of observations from each distribution must be equal to avoid having measurements from one distribution that do not have the relating pair in the second distribution.

Independent samples occur when you cannot reasonably link observations together.

Inference about means from 2 independent samples:

If we are interested in comparing the mean measurement between 2 independent samples we will focus on estimating the difference between the means, $\mu_1 - \mu_2$. With this context, if there is no measurable difference in the means the difference will produce an outcome of 0 and it is reasonable for use to assume the standard deviation of each population in question are equal ($\sigma_1 = \sigma_2$) we can define the method for a $(1 - \alpha)\%$ level of confidence with the following method:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{(v, \alpha/2)}^* s_p \sqrt{\left[\frac{1}{n_1} + \frac{1}{n_2} \right]}$$

Where

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2) - 2}}$$

In this method s_p represents the pooled estimate for the common standard deviation between the distributions. With this method we are assuming the distribution of each sample mean is approximately normal under the CLT and our degrees of freedom for the t-critical value is $n_1 + n_2 - 2$.

If we do not assume the deviations are equal, we would use the individual sample variances in the calculation of the overall standard error of the difference in means. Giving the method:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{(v, \alpha/2)}^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

In this method we must still assume that each sample mean is approximately normal; however, the degrees of freedom

are more complicated to calculate.

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

In general, the results of this calculation will not be an integer and we will need to round down the value for the degrees of freedom of the process. As s_1 and s_2 approach each other in value the overall degrees of freedom approach $n_1 + n_2 - 2$.

For situations where we are working with paired or dependent samples, we can simplify the 2 sample nature of the analysis by focusing on the differences calculated between each measurement of each individual in the study. This effectively makes the paired samples process act as a one sample process on the differences between the measurements.

$$\bar{x}_d \pm t_{(v, \alpha/2)}^* \left\{ s_d / \sqrt{n_d} \right\}$$

For this process, we may simplify the degrees of freedom as $n_d - 1$ and we will operate under the assumption that the sample mean difference behaves approximately normal.

Inferences concerning proportions

Expanding on the one sample case and using a similar point of view as done with 2 sample means we are interested in estimating the difference between the proportions from each distribution, where we must assume that each proportion follows an approximate normal distribution. For this case the method to construct the interval estimate will be:

$$(\hat{p}_1 - \hat{p}_2) \pm z_{(1-\alpha/2)}^* \sqrt{\left[\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2} \right]}$$

In this instance, since the method relies on the z distribution for critical values we do not need to worry about any degrees of freedom calculations.

Inferences concerning variance/standard deviation

Operating under the assumption that $\frac{(n-1)s^2}{\sigma^2}$ behaves as χ^2 with $n-1$ degrees of freedom (provided a normal distribution of the population) we may argue that the behavior of $\frac{s_1^2}{s_2^2}$ would follow the behavior of the F distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ since the distribution of $\frac{\chi_1^2}{\chi_2^2} = F_{(v_1, v_2)}$. Our $(1 - \alpha)\%$ for the ratio of variances is provided by the method:

$$F_{(v_1, v_2, \alpha/2)} \frac{s_1^2}{s_2^2} < \frac{\sigma_1^2}{\sigma_2^2} < F_{(v_1, v_2, 1-\alpha/2)} \frac{s_1^2}{s_2^2}$$

Operating under the assumption that both underlying population distributions are approximately normal. Note: unlike the other 2 sample methods described above, we do not take the difference of the variances since such a difference would be considered a measure of variability itself and variance must be a positive figure. By taking the ratio of the variances we need not be concerned about the possibility of a negative ratio. The "no difference" criteria for this method is 1, as long as the bounds of this interval contain that value we may argue with $(1 - \alpha)\%$ confidence that there is no difference between variances of the distributions.

Next week we will be discussing the 2-sample hypothesis testing approaches.

2-Sample Inference - Hypothesis Testing

If we are interested in comparing the mean measurement between 2 independent samples we are interested in the difference between the means and formulate our null hypothesis around such concepts. In this case, the null hypothesis is of the form $H_0: \mu_1 - \mu_2 = \delta_0$ where δ_0 represents the hypothesized difference between the means on the population level. For general comparisons between mean values this difference is typically assumed to be zero. For this initial case, if we assume equal population variances, we can simplify the test statistic of the testing process with:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2) - 2}}$$

In this method s_p represents the pooled estimate for the common standard deviation between the distributions. With this method we are assuming the distribution of each sample mean is approximately normal under the CLT and our degrees of freedom for the test is $n_1 + n_2 - 2$.

If we do not assume the deviations are equal, we would use the individual sample variances in the calculation of the overall standard error of the difference in means. Giving the method:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

In this method we must still assume that each sample mean is approximately normal; however, the degrees of freedom are more complicated to calculate.

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

For situations where we are working with paired or dependent samples, we modify the hypothesis format provided above in favor of one that focuses on the true mean difference of the measurements. This hypothesis is of the form: $H_0: \mu_d = \delta_0$ and our method for computing the paired t-statistic is:

$$t = \frac{\bar{x}_d - \mu_d}{s_d / \sqrt{n}}$$

For this process, we may simplify the degrees of freedom as $n_d - 1$ and we will operate under the assumption that the sample mean difference behaves approximately normal.

Inferences concerning proportions

When performing testing on 2 samples focusing on categorical data types, the hypotheses of interest are typically of the form: $H_0: p_1 - p_2 = \delta_0$. Unlike interval estimates for 2 sample proportions where we have one primary method for construction of the confidence bounds, for the cases where we assume the difference between the proportions is equal to zero, it is reasonable to assume that each proportion is equal. This is to say, if $H_0: p_1 - p_2 = 0$ then it is reasonable to assume $p_1 = p_2$ and operating on this assumption from the null hypothesis produces a method for the test statistic:

$$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}_c(1 - \hat{p}_c) \left[\frac{1}{n_1} + \frac{1}{n_2} \right]}}$$

Where $\hat{p}_c$ is called the **pooled estimate for the proportion** since under the assumption $p_1 = p_2$ our proportions should be equal to some overall proportion, p . This pooled estimate is meant to serve as a point estimate for the common proportion of the problem. The pooled estimate can be calculated with the formula:

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

For general differences between proportions under the null hypothesis, the test statistic takes the form:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - \delta_0}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}}$$

Inferences concerning variance/standard deviation

Continuing with assumptions that the populations sampled from are normally distributed we may continue to assume that the distribution of one Chi-square distribution to another follows a value defined by the F distribution which carries degrees of freedom defined by each of the individual degrees of freedom of the distributions that define the ratio. For similar reasons outlined in our interval estimate methods, we want to avoid associating variance or standard deviation with a difference. Instead, we look at the ratio between the variance or deviations under our null hypothesis, taking the form: $H_0: \frac{\sigma_1^2}{\sigma_2^2} = 1$ allowing for a general comparison between the variances of the distributions. The test statistic for this case can be calculated as follows:

$$F = \frac{s_1^2}{s_2^2}$$

Which produces a value of the F distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom.